Module 11: Genomics and Biotechnology - Key Points

Learning Objectives

By the end of this module, you should be able to:

1. Define genome and genomics.
2. Explain why PCR amplification is useful.
3. Interpret basic gel electrophoresis fragment patterns.
4. Describe what genetic engineering tools can do.
5. Evaluate biotechnology claims using evidence and ethical considerations.

Topics

- Genomes as data
- PCR and amplification
- Gel electrophoresis and comparison
- Genetic engineering tools
- Biotechnology decisions and ethics

Key Terms to Know

- **Genome** - Complete DNA information of an organism or sample.
- **Genomics** - Study of whole genomes and their patterns.
- **PCR** - Technique for amplifying selected DNA sequences.
- **Primer** - Short DNA sequence that starts PCR copying.
- **Gel electrophoresis** - Method that separates DNA fragments by size.
- **CRISPR** - Gene-editing system adapted from bacterial defense.
- **GMO** - Organism with genetic material altered by biotechnology.
- **Bioethics** - Reasoning about benefits, harms, rights, and responsibilities in biology.

Core Contents

1. A genome is the full DNA information set of an organism or sample.
2. PCR amplifies chosen DNA regions so they can be analyzed.
3. Gel electrophoresis separates DNA fragments by size.
4. Biotechnology tools can copy, edit, compare, or move genetic information.
5. Applications require evidence, risk reasoning, and ethical judgment.

Study Tips

1. Start with the module's central claim before memorizing vocabulary.
2. Use the same-numbered lab as the evidence check for the module.
3. Explain one mechanism in words, then redraw it as a simple diagram.
4. Answer retrieval questions without notes, then revise with the terms list.

Connected Lab

- `lab-11_genomics-biotechnology.md`

Generated Visuals

- Module 11: Genomics and Biotechnology Evidence Map
- Module 11: Genomics and Biotechnology Reasoning Sequence
- Module 11: Genomics and Biotechnology Retrieval and Lab Check